

Fejér–heat generators and Lipschitz control for the Weil quadratic functional

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Abstract

We record two basic analytic properties of the Weil quadratic functional Q associated with the Riemann zeta function in the Guinand–Weil normalization. On the frequency side, Q acts on even, compactly supported test functions Φ with $\text{supp } \Phi \subset [-K, K]$ and decomposes as an Archimedean term minus an explicit prime contribution.

For each $K > 0$ we introduce a Fejér–heat cone $\mathcal{G}_K$ of frequency windows obtained by convolving classical Fejér kernels with Gaussian heat kernels and taking finite nonnegative linear combinations of even translates. We prove that $\mathcal{G}_K$ is dense in the cone $C_{\text{even}}^+([-K, K])$ of continuous, even, nonnegative functions with respect to the supremum norm. Furthermore, we show that on each window $[-K, K]$ the functional Q is Lipschitz continuous with respect to the supremum norm, with an explicit Lipschitz constant of the form

$$L_Q(K) = L_{\text{arch}}(K) + L_{\text{prime}}(K),$$

where $L_{\text{arch}}(K)$ arises from the Archimedean contribution and $L_{\text{prime}}(K)$ from a truncated sum over primes and prime powers.

These results yield a compact–by–compact approximation and continuity framework for the localized Weil functional in Guinand–Weil normalization. They are proved independently of any global positivity assumptions and should be viewed as a self-contained technical note providing tools that may be useful in operator–theoretic approaches to the Weil positivity criterion.

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1 Introduction

The Weil positivity criterion is one of the classical reformulations of the Riemann Hypothesis. In its standard form it associates to the Riemann zeta function a quadratic functional Q on a suitable cone of even test functions Φ and asserts that

$$\text{Riemann Hypothesis} \iff Q(\Phi) \geq 0 \quad \text{for all admissible } \Phi.$$

In this way the location of the nontrivial zeros—whether they all lie on the critical line or not—is encoded in the positivity of a single quadratic form. This point of view goes back to Weil’s seminal work on explicit formulas for global fields [11, 12] and has since appeared in various guises in analytic number theory and in operator–theoretic approaches to zeta and L –functions. In this note we restrict attention to unconditional local analytic properties of the associated Weil functional.

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We work throughout in the Guinand–Weil normalization of the explicit formula and adopt a localized, compact-by-compact perspective on the Weil functional Q . On the frequency side we consider even, compactly supported test functions Φ with $\text{supp } \Phi \subset [-K, K]$, and we view Q as a functional on the corresponding cone of test functions. In this setting Q decomposes naturally as

$$Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi),$$

where Q_{arch} is an Archimedean contribution involving the gamma factor and Q_{prime} is an explicit prime contribution built from the von Mangoldt function. The precise definitions are recalled in section 3.

The purpose of this note is not to address the global positivity problem for Q , but to isolate two basic analytic tools on each compact window $[-K, K]$: a flexible dense cone of frequency generators and a quantitative supremum–norm Lipschitz bound for Q . The note should be viewed as a self-contained technical contribution; the results are formulated so as to be directly usable in operator–theoretic treatments of the Weil criterion developed elsewhere.

Fejér–heat generators on compact windows

For a fixed $K > 0$, let $C_{\text{even}}^+([-K, K])$ denote the cone of continuous, even, nonnegative functions on the interval $[-K, K]$. Motivated by applications to Toeplitz operators and reproducing kernel Hilbert spaces (RKHS), we introduce a cone of “elementary windows” obtained by convolving classical Fejér kernels with Gaussian heat kernels. More precisely, we consider functions of the form

$$\omega_{B,t} = F_B * \rho_t,$$

where F_B is a Fejér kernel of order B and ρ_t is a one–dimensional heat kernel at time $t > 0$. The Fejér–heat windows $\omega_{B,t}$ are smooth positive functions on all of $\mathbb{R}$; we restrict them to $[-K, K]$ when forming the generator cone $\mathcal{G}_K$. The cone $\mathcal{G}_K$ consists of finite nonnegative linear combinations of such restricted windows and their even symmetrizations.

Our first basic result shows that these Fejér–heat windows generate a dense cone on each compact interval $[-K, K]$.

Theorem 1.1 (Fejér–heat density on $[-K, K]$). *Let $K > 0$. Let $\mathcal{G}_K$ be the cone generated by finite nonnegative linear combinations of even Fejér–heat windows restricted to $[-K, K]$. Then $\mathcal{G}_K$ is dense in $C_{\text{even}}^+([-K, K])$ with respect to the supremum norm. In other words, for every $\psi \in C_{\text{even}}^+([-K, K])$ and every $\varepsilon > 0$ there exists $g \in \mathcal{G}_K$ such that*

$$\|\psi - g\|_{\infty} < \varepsilon.$$

This result provides a flexible family of generators for the frequency cones that is particularly suited to the operator–theoretic constructions developed elsewhere. The proof is based on classical approximation properties of Fejér kernels and heat kernels, combined with a cone–preserving symmetrization argument. The full details are given in section 4.

Lipschitz control of the Weil functional

The second ingredient we develop is a quantitative continuity estimate for the Weil functional Q on each compact window. On the frequency side it is natural to equip our spaces of test functions with the supremum norm. We show that, after fixing $K > 0$, the functional Q is Lipschitz continuous on the corresponding frequency cone, with an explicit Lipschitz constant depending on K .

Theorem 1.2 (Lipschitz continuity of Q on $[-K, K]$). *Let $K > 0$ be fixed. Then there exists a finite constant $L_Q(K) > 0$ such that for any two admissible test functions $\Phi_1, \Phi_2 \in C_{c,\text{even}}^{\infty}$ (see section 2) we have*

$$|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_{\infty}.$$

Moreover, the constant $L_Q(K)$ admits an explicit decomposition

$$L_Q(K) = L_{\text{arch}}(K) + L_{\text{prime}}(K) = \int_{-K}^K |a^*(\xi)| d\xi + 2 \sum_{2 \leq n \leq e^{2\pi K}} \frac{\Lambda(n)}{\sqrt{n}},$$

where $L_{\text{arch}}(K)$ arises from the Archimedean density a^* and $L_{\text{prime}}(K)$ is a finite sum over primes and prime powers up to $e^{2\pi K}$ weighted by the von Mangoldt function.

The proof of theorem 1.2 is elementary: since test functions $\Phi \in C_{c,\text{even}K}^\infty$ are compactly supported in $[-K, K]$, the Archimedean integral reduces to a finite integral and the prime sum becomes a finite sum. The resulting bounds are completely explicit. The details are presented in section 5.

Context and motivation

Theorems 1.1 and 1.2 may serve as building blocks for more global operator–theoretic treatments of the Weil functional. On each compact window $[-K, K]$, the density of Fejér–heat generators and the Lipschitz control of Q provide an approximation and stability framework well adapted to Toeplitz operators and reproducing kernel Hilbert spaces. These tools can be combined with Toeplitz–symbol lower bounds and reproducing kernel estimates for the prime contribution to formulate operator–theoretic frameworks for studying the Weil positivity criterion on the full Weil cone. The present article is logically independent of any global positivity claims: all results here are unconditional analytic statements about the localized Weil functional in the Guinand–Weil normalization, and nothing in the proofs uses conjectural information about zeros.

In Weil’s adelic reformulation of the explicit formulas [12], the positivity problem is encoded in a distribution Δ acting on test functions on the idèle class group, and the positivity of Δ is equivalent to the Riemann Hypothesis together with Artin’s conjecture for the corresponding L -functions. Our quadratic functional Q is the one–dimensional, frequency-side avatar of this distribution, specialized to the Riemann zeta function in the Guinand–Weil normalization.

Related work. The use of explicit formulas of Guinand–Weil type to encode the zero distribution of $\zeta(s)$ and more general L -functions goes back to the classical works of Guinand and Weil [7, 11, 12], and is systematically developed in standard references such as Edwards and Iwaniec–Kowalski [4, 8]. In that framework the positivity problem for suitable quadratic forms attached to ζ is usually treated either globally, at the level of adelic distributions, or via test functions adapted to particular analytic inequalities. The present note takes a more localized, operator–oriented point of view: we restrict to compact frequency windows, introduce a Fejér–heat cone of generators tailored to Toeplitz and reproducing kernel Hilbert space methods, and establish quantitative Lipschitz control of the Weil functional on each window. As far as we are aware, this specific combination of Fejér–heat generators and Lipschitz bounds for the Guinand–Weil functional has not been formulated explicitly in this localized setting.

2 Preliminaries and notation

In this section we fix the basic function spaces, norms and transform conventions used throughout the paper. We work on the real line $\mathbb{R}$ on the frequency side, and on compact windows $[-K, K]$ with $K > 0$ fixed.

2.1 Function spaces and norms

For $K > 0$ we write $C([-K, K])$ for the space of continuous complex-valued functions on the compact interval $[-K, K]$, equipped with the supremum norm

$$\|\psi\|_\infty := \sup_{x \in [-K, K]} |\psi(x)|.$$

We denote by $C_{\text{even}}([-K, K])$ the subspace of *even* functions,

$$C_{\text{even}}([-K, K]) := \{\psi \in C([-K, K]) : \psi(-x) = \psi(x) \text{ for all } x \in [-K, K]\},$$

and by $C_{\text{even}}^+([-K, K])$ the associated *cone of nonnegative even functions*,

$$C_{\text{even}}^+([-K, K]) := \{\psi \in C_{\text{even}}([-K, K]) : \psi(x) \geq 0 \text{ for all } x \in [-K, K]\}.$$

Unless explicitly stated otherwise, all function spaces on $[-K, K]$ will be equipped with the supremum norm $\|\cdot\|_\infty$.

On the frequency side we work with *admissible test functions* on a compact window. For $K > 0$ we define the space of *even compactly supported test functions*

$$C_{c,\text{even}K}^\infty := \{\Phi \in C^\infty(\mathbb{R}) : \text{supp } \Phi \subset [-K, K], \Phi(-\xi) = \Phi(\xi) \text{ for all } \xi\}, \quad (1)$$

i.e., the space of even, smooth functions on $\mathbb{R}$ whose support is contained in the compact window $[-K, K]$. Every $\Phi \in C_{c,\text{even}K}^\infty$ is continuous, even, and vanishes identically outside $[-K, K]$. When we say “let Φ be an admissible test function on $[-K, K]$ ” we mean $\Phi \in C_{c,\text{even}K}^\infty$.

Remark 2.1 (Choice of test class). Working with compactly supported test functions is a deliberate simplification that makes all convergence issues trivial: for $\Phi \in C_{c,\text{even}K}^\infty$ the Archimedean integral reduces to an integral over $[-K, K]$, and the prime sum becomes a finite sum over those n with $\xi_n = (\log n)/(2\pi) \in [-K, K]$. This is the natural setting for the “compact-by-compact” approach to the Weil criterion adopted here. For more general test classes (e.g., Paley–Wiener functions arising as Fourier transforms of compactly supported distributions), see [7, 12].

2.2 Fourier transform conventions

We use the following *Fourier transform* convention on $\mathbb{R}$:

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(t) e^{-2\pi i t \xi} dt, \quad \phi(t) = \int_{\mathbb{R}} \widehat{\phi}(\xi) e^{2\pi i t \xi} d\xi, \quad (2)$$

whenever the integrals make sense.

In the rest of the paper we work primarily on the frequency side, and we will often refer to $\Phi(\xi)$ as a *frequency window*. When needed, we identify Φ with a time-side test function g via (2):

$$\Phi(\xi) = \widehat{g}(\xi), \quad g(t) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i t \xi} d\xi.$$

We will always take Φ to be even, so that the corresponding g is real-valued.

2.3 The Fejér kernel

We use a continuous analogue of the classical Fejér kernel, adapted to the real line rather than the circle. For an integer $B \geq 1$ we define the *Fejér kernel on $\mathbb{R}$* by

$$F_B(x) = B \left(\frac{\sin(\pi B x)}{\pi B x} \right)^2, \quad x \in \mathbb{R}, x \neq 0, \quad (3)$$

with the continuous extension $F_B(0) = B$ at the origin. Equivalently, $F_B(x) = B \operatorname{sinc}^2(Bx)$ where $\operatorname{sinc}(u) := \sin(\pi u)/(\pi u)$ is the normalized sinc function.

The kernel F_B is smooth, even, and nonnegative on $\mathbb{R}$, with $F_B(x) > 0$ for $|x| < 1/B$. It belongs to $L^1(\mathbb{R})$ and has total mass

$$\int_{\mathbb{R}} F_B(x) dx = 1. \quad (4)$$

The Fourier transform of F_B is the triangular window

$$\widehat{F}_B(\xi) = \begin{cases} 1 - |\xi|/B, & |\xi| \leq B, \\ 0, & |\xi| > B, \end{cases} \quad (5)$$

which has compact support in $[-B, B]$ and satisfies $\widehat{F}_B(0) = 1$.

As $B \rightarrow \infty$, the family $(F_B)_{B \geq 1}$ forms a classical positive approximate identity on $\mathbb{R}$; see, e.g., [5, 9, 13] for background on Fejér kernels and harmonic analysis. For every bounded, uniformly continuous function $f: \mathbb{R} \rightarrow \mathbb{C}$ one has

$$\|f * F_B - f\|_{\infty} \rightarrow 0 \quad (B \rightarrow \infty),$$

where $*$ denotes convolution on $\mathbb{R}$.

2.4 The heat kernel

We also use the one-dimensional *heat kernel* (Gaussian kernel) on $\mathbb{R}$, defined for $t > 0$ by

$$\rho_t(x) = \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{x^2}{4t}\right), \quad x \in \mathbb{R}. \quad (6)$$

The kernel ρ_t is smooth, even, and strictly positive, with total mass

$$\int_{\mathbb{R}} \rho_t(x) dx = 1.$$

As $t \rightarrow 0^+$, the family $(\rho_t)_{t>0}$ forms a standard approximate identity on $\mathbb{R}$: for every continuous function f on $\mathbb{R}$ with at most polynomial growth at infinity, the convolutions $f * \rho_t$ converge locally uniformly to f . For background on heat kernels and their analytic properties, see [3, 6].

2.5 Lipschitz continuity

We record the basic notion of *Lipschitz functional* that will be used for the Weil functional Q on compact frequency windows.

Definition 2.2 (Lipschitz functional). Let $(X, \|\cdot\|)$ be a normed space and let $F: X \rightarrow \mathbb{C}$ be a functional. We say that F is *Lipschitz continuous* with Lipschitz constant $L \geq 0$ if

$$|F(x) - F(y)| \leq L \|x - y\|$$

for all $x, y \in X$.

In later sections we will apply this notion to the Weil functional Q restricted to admissible test functions $\Phi \in C_{c,\text{even}}^{\infty} K$, equipped with the supremum norm $\|\cdot\|_{\infty}$.

3 The Weil quadratic functional in Guinand–Weil normalization

In this section we recall the Guinand–Weil explicit formula for the Riemann zeta function and use it to define the associated Weil quadratic functional Q on even, compactly supported test functions. We then decompose Q into its Archimedean and prime contributions.

3.1 The Guinand–Weil explicit formula

We start from the Guinand–Weil explicit formula for the Riemann zeta function. Let $g : \mathbb{R} \rightarrow \mathbb{C}$ be an even Schwartz function such that its Fourier transform $\widehat{g}$ is compactly supported. Then the Guinand–Weil explicit formula (see [7, 11, 12]) can be written schematically as

$$\sum_{\rho} \widehat{g}\left(\frac{\rho - \frac{1}{2}}{i}\right) = (\text{Archimedean term}) - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n), \quad (7)$$

where the sum on the left runs over the nontrivial zeros ρ of $\zeta(s)$ and the Archimedean term involves the logarithmic derivative of the gamma factor.

We do not need the detailed form of the Archimedean contribution here; what matters for us is that, after a change of variables to the frequency side, the right-hand side of (7) can be written as a linear functional of a frequency window Φ with an explicit Archimedean density and an explicit prime sum.

3.2 Definition of the Weil functional

We work on the frequency axis with coordinate $\xi = \eta/(2\pi)$. Following the conventions of [7, 11, 12], we define the *Archimedean density*

$$a(\xi) = \log \pi - \Re \psi\left(\frac{1}{4} + i\pi\xi\right), \quad a^*(\xi) = 2\pi a(\xi), \quad (8)$$

where $\psi = \Gamma'/\Gamma$ is the digamma function.

Prime nodes are placed at

$$\xi_n = \frac{\log n}{2\pi}, \quad n \geq 2, \quad (9)$$

and we use the one-sided *Weil weights*

$$w_Q(n) = \frac{2\Lambda(n)}{\sqrt{n}}, \quad (10)$$

where Λ is the von Mangoldt function.

Definition 3.1 (Weil quadratic functional). Let $\Phi \in C_{c,\text{even}K}^\infty$ be an admissible test function (see section 2). The *Weil quadratic functional* in Guinand–Weil normalization is the linear functional

$$Q(\Phi) = \int_{-K}^K a^*(\xi) \Phi(\xi) d\xi - \sum_{\xi_n \in [-K, K]} w_Q(n) \Phi(\xi_n), \quad (11)$$

with a^* , ξ_n and $w_Q(n)$ as in (8)–(10).

Strictly speaking, Q is linear in Φ ; we call it “quadratic” because in applications to the Weil criterion one typically takes Φ to be a self-convolution $\Phi = g * \tilde{g}$ of a test function g , so that $Q(\Phi)$ becomes a quadratic form in g .

Remark 3.2 (Well-posedness). For $\Phi \in C_{c,\text{even}K}^\infty$ the functional $Q(\Phi)$ is trivially well-defined: since $\text{supp } \Phi \subset [-K, K]$, the Archimedean integral reduces to a finite integral over $[-K, K]$, and the prime sum is a finite sum over those $n \geq 2$ with $\xi_n = (\log n)/(2\pi) \in [-K, K]$, i.e., over $2 \leq n \leq e^{2\pi K}$.

3.3 Archimedean and prime decomposition

It is convenient to split Q into its Archimedean and prime parts.

Proposition 3.3 (Decomposition of Q). *For any admissible test function Φ as in Definition 3.1, the Weil functional decomposes as*

$$Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi),$$

where

$$Q_{\text{arch}}(\Phi) = \int_{\mathbb{R}} a^*(\xi) \Phi(\xi) d\xi, \quad (12)$$

$$Q_{\text{prime}}(\Phi) = \sum_{n \geq 2} w_Q(n) \Phi(\xi_n). \quad (13)$$

Proof. This is a restatement of Definition 3.1 with the Archimedean integral and prime sum written as separate functionals. $\square$

Remark 3.4 (Normalization and link to Weil’s explicit formulas). The normalization adopted here is compatible with the Guinand–Weil form of the explicit formula for the Riemann zeta function [7, 11, 12]. In Weil’s adelic framework, the positivity problem is encoded in a distribution Δ acting on test functions on the idèle class group, and the positivity of Δ is equivalent to the Riemann Hypothesis together with Artin’s conjecture for the corresponding L -functions. Our functional Q is the one-dimensional frequency-side avatar of this distribution, specialized to the Riemann zeta function in the Guinand–Weil normalization. We define Q using the standard Archimedean density and Weil weights from the explicit formula; on the Paley–Wiener test class this coincides with the classical Weil functional, and (11) should be viewed as its localized extension to compactly supported frequency windows. This remark is included for context only; the results of the present note do not rely on the positivity of Δ or on any unproved conjectures.

Remark 3.5 (Relation to Paley–Wiener spaces). In operator-theoretic treatments of the Weil criterion one often works with Paley–Wiener test functions, i.e., functions $g \in C_c^\infty([-T, T])$ on the time side and their Fourier transforms $\Phi = \widehat{g}$. By the Paley–Wiener theorem, such Φ are entire functions and do *not* have compact support on the frequency side; thus they do not belong to $C_{c,\text{even}K}^\infty$ as defined in (1). Nonetheless, the density and Lipschitz results of this paper remain useful for the operator-theoretic setting: the space $C_{c,\text{even}K}^\infty$ is dense in natural topologies on Paley–Wiener functions restricted to $[-K, K]$, so the quantitative estimates established here extend by continuity to broader test function classes.

Remark 3.6 (Notation for weights). We use $w_Q(n)$ for the Weil weights appearing in the definition of Q . In later sections we will also encounter other weight systems (for example, in the Lipschitz estimates); these will be clearly distinguished by subscripts.

4 Fejér–heat generators and density on $[-K, K]$

In this section we introduce a cone of Fejér–heat frequency windows and prove that it is dense in the cone of continuous, even, nonnegative functions on each compact interval $[-K, K]$. Throughout we fix $K > 0$.

4.1 The Fejér–heat convolution

Definition 4.1 (Fejér–heat window). Let $B \geq 1$ be an integer and let $t > 0$. Let $F_B: \mathbb{R} \rightarrow \mathbb{R}$ be the one-dimensional Fejér kernel of order B (see section 2), and let $\rho_t: \mathbb{R} \rightarrow \mathbb{R}$ be the

one-dimensional heat kernel at time t (see [section 2](#)). We define the *Fejér-heat window* of order (B, t) by

$$\omega_{B,t} = F_B * \rho_t,$$

where $*$ denotes convolution on $\mathbb{R}$.

By construction, F_B is smooth, nonnegative and even, while ρ_t is smooth, strictly positive and even; both belong to $L^1(\mathbb{R})$ with total mass 1 (see [section 2](#)). As a convolution of a nonnegative function with a strictly positive one, $\omega_{B,t}$ is itself strictly positive. Moreover, the families $\{F_B\}_{B \geq 1}$ and $\{\rho_t\}_{t > 0}$ form positive approximate identities on $\mathbb{R}$: for every bounded, uniformly continuous function $f: \mathbb{R} \rightarrow \mathbb{C}$ one has

$$\|f * F_B - f\|_\infty \longrightarrow 0 \quad (B \rightarrow \infty), \quad \|f * \rho_t - f\|_\infty \longrightarrow 0 \quad (t \rightarrow 0^+).$$

4.2 Basic properties of Fejér-heat windows

Lemma 4.2 (Basic properties of Fejér-heat windows). *For every integer $B \geq 1$ and $t > 0$ the Fejér-heat window $\omega_{B,t}$ satisfies:*

- (i) $\omega_{B,t}(x) \geq 0$ for all $x \in \mathbb{R}$ (positivity);
- (ii) $\omega_{B,t}(-x) = \omega_{B,t}(x)$ for all $x \in \mathbb{R}$ (evenness);
- (iii) $\int_{\mathbb{R}} \omega_{B,t}(x) dx = 1$ (normalization).

Proof. By definition,

$$\omega_{B,t}(x) = (F_B * \rho_t)(x) = \int_{\mathbb{R}} F_B(y) \rho_t(x - y) dy.$$

(i) Since $F_B(y) \geq 0$ and $\rho_t(z) \geq 0$ for all $y, z \in \mathbb{R}$, the integrand is nonnegative, hence $\omega_{B,t}(x) \geq 0$ for all x .

(ii) Using that both F_B and ρ_t are even functions, we obtain

$$\omega_{B,t}(-x) = \int_{\mathbb{R}} F_B(y) \rho_t(-x - y) dy = \int_{\mathbb{R}} F_B(-y) \rho_t(x - y) dy = \int_{\mathbb{R}} F_B(y) \rho_t(x - y) dy = \omega_{B,t}(x),$$

where we changed variables $y \mapsto -y$ and used evenness.

(iii) Using Fubini's theorem and the normalization $\int_{\mathbb{R}} F_B = \int_{\mathbb{R}} \rho_t = 1$, we have

$$\int_{\mathbb{R}} \omega_{B,t}(x) dx = \int_{\mathbb{R}} \int_{\mathbb{R}} F_B(y) \rho_t(x - y) dy dx = \left(\int_{\mathbb{R}} F_B(y) dy \right) \left(\int_{\mathbb{R}} \rho_t(u) du \right) = 1 \cdot 1 = 1.$$

This proves the three properties. □

4.3 The Fejér-heat cone

In applications we want to work with *even* windows on the frequency axis. We obtain these by symmetrizing translates of Fejér-heat windows.

Definition 4.3 (Even Fejér-heat windows). Let $B \geq 1$ be an integer, $t > 0$ and $x_0 \in \mathbb{R}$. The associated *even Fejér-heat window centered at x_0* is the function

$$\Omega_{B,t,x_0}(x) = \frac{1}{2}(\omega_{B,t}(x - x_0) + \omega_{B,t}(x + x_0)), \quad x \in \mathbb{R}.$$

By [theorem 4.2](#) and the definition, each Ω_{B,t,x_0} is continuous, nonnegative and even, with $\int_{\mathbb{R}} \Omega_{B,t,x_0}(x) dx = 1$.

Definition 4.4 (Fejér–heat cone). For $K > 0$ we define the *Fejér–heat cone* $\mathcal{G}_K$ to be the set of all functions on $[-K, K]$ that can be written in the form

$$g(x) = \sum_{j=1}^N c_j \Omega_{B_j, t_j, x_j}(x), \quad x \in [-K, K],$$

for some integer $N \geq 1$, integers $B_j \geq 1$, parameters $t_j > 0$, centers $x_j \in \mathbb{R}$, and coefficients $c_j \geq 0$. Equivalently, $\mathcal{G}_K$ is the cone generated by finite nonnegative linear combinations of even Fejér–heat windows restricted to $[-K, K]$.

By construction, every $g \in \mathcal{G}_K$ is continuous, even and nonnegative on $[-K, K]$.

4.4 Density theorem

We now show that $\mathcal{G}_K$ is dense in the cone of continuous, even, nonnegative functions on $[-K, K]$ with respect to the supremum norm. This is the localized form of theorem 1.1 from the introduction.

Theorem 4.5 (Fejér–heat density). *Let $K > 0$. Then the cone $\mathcal{G}_K$ is dense in $C_{\text{even}}^+([-K, K])$ with respect to the supremum norm $\|\cdot\|_\infty$. More precisely, for every $\psi \in C_{\text{even}}^+([-K, K])$ and every $\varepsilon > 0$ there exists $g \in \mathcal{G}_K$ such that*

$$\|\psi - g\|_\infty < \varepsilon.$$

Proof. Fix $K > 0$, $\psi \in C_{\text{even}}^+([-K, K])$ and $\varepsilon > 0$. Throughout this proof, all supremum norms $\|\cdot\|_\infty$ without explicit domain annotation are taken over $[-K, K]$.

Step 1: even extension to the real line. Define an even, bounded, continuous extension $\tilde{\psi}: \mathbb{R} \rightarrow [0, \infty)$ by

$$\tilde{\psi}(x) = \begin{cases} \psi(x), & |x| \leq K, \\ \psi(K), & |x| > K. \end{cases}$$

Since ψ is continuous and even on $[-K, K]$, the function $\tilde{\psi}$ is continuous, even and nonnegative on $\mathbb{R}$. Let $M := \|\tilde{\psi}\|_\infty < \infty$.

Step 2: approximation by Fejér and heat kernels. By the approximate identity property of the heat kernels ρ_t , there exists $t_0 > 0$ such that for all $0 < t \leq t_0$ we have

$$\|\tilde{\psi} * \rho_t - \tilde{\psi}\|_{L^\infty([-K, K])} < \frac{\varepsilon}{4}. \quad (14)$$

Fix one such $t \in (0, t_0]$ and set

$$f(x) := (\tilde{\psi} * \rho_t)(x), \quad x \in \mathbb{R}.$$

Then f is continuous, bounded, even and nonnegative.

Similarly, by the approximate identity property of the Fejér kernels F_B , there exists $B_0 > 0$ such that for all $B \geq B_0$ we have

$$\|f * F_B - f\|_\infty < \frac{\varepsilon}{4}. \quad (15)$$

For such B we note that, by associativity and commutativity of convolution,

$$(\tilde{\psi} * \omega_{B,t})(x) = (\tilde{\psi} * (F_B * \rho_t))(x) = ((\tilde{\psi} * \rho_t) * F_B)(x) = (f * F_B)(x).$$

Combining (14) and (15), we obtain for all $x \in [-K, K]$:

$$|\tilde{\psi} * \omega_{B,t}(x) - \psi(x)| \leq |\tilde{\psi} * \omega_{B,t}(x) - f(x)| + |f(x) - \tilde{\psi}(x)| < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}.$$

In other words,

$$\|\tilde{\psi} * \omega_{B,t} - \psi\|_{L^\infty([-K,K])} < \frac{\varepsilon}{2}. \quad (16)$$

Step 3: Riemann sum approximation by finite mixtures. Set $\omega := \omega_{B,t}$ for the fixed parameters from Step 2. By [theorem 4.2](#) we have $\omega \geq 0$ and $\int_{\mathbb{R}} \omega = 1$.

Let $R > K$ be a parameter to be chosen. For $x \in [-K, K]$ we write

$$(\tilde{\psi} * \omega)(x) = \int_{\mathbb{R}} \tilde{\psi}(y) \omega(x - y) dy = \int_{-R}^R \tilde{\psi}(y) \omega(x - y) dy + \int_{|y| > R} \tilde{\psi}(y) \omega(x - y) dy.$$

Since $\int_{\mathbb{R}} \omega = 1$ and ω is integrable, we can choose R large enough so that

$$\int_{|u| > R-K} \omega(u) du < \frac{\varepsilon}{4M}.$$

Then for all $x \in [-K, K]$ we have

$$\left| \int_{|y| > R} \tilde{\psi}(y) \omega(x - y) dy \right| \leq M \int_{|y| > R} \omega(x - y) dy = M \int_{|u+x| > R} \omega(u) du \leq M \int_{|u| > R-K} \omega(u) du < \frac{\varepsilon}{4}.$$

Thus the tail outside $[-R, R]$ contributes at most $\varepsilon/4$ in supremum norm on $[-K, K]$.

On the compact set $[-K, K] \times [-R, R]$ the function

$$\Phi(x, y) := \tilde{\psi}(y) \omega(x - y)$$

is continuous and hence uniformly continuous. Therefore there exists $\eta > 0$ such that

$$|y - y'| < \eta \implies |\Phi(x, y) - \Phi(x, y')| < \frac{\varepsilon}{4 \cdot 2R} \quad \text{for all } x \in [-K, K], y, y' \in [-R, R].$$

Partition the interval $[-R, R]$ into subintervals $I_j = [y_j - \Delta_j/2, y_j + \Delta_j/2]$ of length $0 < \Delta_j \leq \eta$, with midpoints y_j , $j = 1, \dots, N$, so that $\sum_j \Delta_j = 2R$. Define coefficients $c_j := \tilde{\psi}(y_j) \Delta_j \geq 0$, and set

$$g_0(x) := \sum_{j=1}^N c_j \omega(x - y_j), \quad x \in \mathbb{R}.$$

By construction g_0 is a finite nonnegative linear combination of translates of ω .

For $x \in [-K, K]$ we have

$$\int_{-R}^R \tilde{\psi}(y) \omega(x - y) dy = \sum_{j=1}^N \int_{I_j} \Phi(x, y) dy,$$

and hence

$$\left| \int_{-R}^R \tilde{\psi}(y) \omega(x - y) dy - g_0(x) \right| = \left| \sum_{j=1}^N \int_{I_j} (\Phi(x, y) - \Phi(x, y_j)) dy \right| \leq \sum_{j=1}^N \Delta_j \cdot \frac{\varepsilon}{4 \cdot 2R} = \frac{\varepsilon}{4}.$$

Combining this with the tail estimate, we obtain

$$\|\tilde{\psi} * \omega - g_0\|_{L^\infty([-K,K])} < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}.$$

Step 4: even symmetrization and membership in $\mathcal{G}_K$. By definition, g_0 is a finite nonnegative linear combination of translates of ω . We now enforce evenness by symmetrization. Define

$$g(x) := \frac{1}{2}(g_0(x) + g_0(-x)), \quad x \in \mathbb{R}.$$

Then g is even and nonnegative, and on $[-K, K]$ we have

$$g(x) = \sum_{j=1}^N c_j \frac{1}{2} (\omega(x - y_j) + \omega(x + y_j)) = \sum_{j=1}^N c_j \Omega_{B,t,x_j}(x)$$

with $x_j := y_j$, so that $g|_{[-K,K]} \in \mathcal{G}_K$ by [theorem 4.4](#).

Since ψ is even, the symmetrization cannot increase the distance to ψ : for every $x \in [-K, K]$,

$$|\psi(x) - g(x)| = \left| \psi(x) - \frac{1}{2}(g_0(x) + g_0(-x)) \right| \leq \frac{1}{2} (|\psi(x) - g_0(x)| + |\psi(-x) - g_0(-x)|) \leq \|\psi - g_0\|_\infty.$$

Therefore

$$\|\psi - g\|_\infty \leq \|\psi - g_0\|_\infty \leq \|\psi - \tilde{\psi} * \omega\|_\infty + \|\tilde{\psi} * \omega - g_0\|_\infty < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where we used [\(16\)](#) and the estimate from Step 3.

Finally, $g|_{[-K,K]}$ belongs to $\mathcal{G}_K$ and satisfies $\|\psi - g\|_\infty < \varepsilon$, which proves the theorem. $\square$

5 Lipschitz control of Q on compact frequency windows

In this section we prove [theorem 1.2](#), establishing that the Weil quadratic functional Q is Lipschitz continuous on each compact frequency window $[-K, K]$ when restricted to admissible test functions $\Phi \in C_{c,\text{even}}^\infty K$. Throughout we fix $K > 0$.

Recall from [proposition 3.3](#) that for any admissible test function $\Phi \in C_{c,\text{even}}^\infty K$ we have the decomposition

$$Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi),$$

where Q_{arch} is the Archimedean contribution coming from the gamma factor and Q_{prime} is the prime contribution built from the von Mangoldt function. We will prove Lipschitz bounds for each of these pieces separately and then combine them.

5.1 Lipschitz estimate for the Archimedean part

We begin with the Archimedean part Q_{arch} . Recall from [section 3](#) that in the Guinand–Weil normalization the Archimedean contribution can be written in the form

$$Q_{\text{arch}}(\Phi) = \int_{-K}^K a^*(\xi) \Phi(\xi) d\xi, \tag{17}$$

where $a^*(\xi)$ is the symmetrized Archimedean density built from the digamma function ψ ,

$$a^*(\xi) = 2\pi \left(\log \pi - \Re \psi \left(\frac{1}{4} + i\pi \xi \right) \right),$$

cf. [section 3](#). Standard asymptotics for the digamma function on vertical lines (see, for example, [\[4, Chap. 1\]](#) or [\[8, Sec. 5.1\]](#)) show that

$$a^*(\xi) = O(\log(2 + |\xi|)), \quad |\xi| \rightarrow \infty, \tag{18}$$

and in particular a^* is smooth and locally integrable on $\mathbb{R}$.

Proposition 5.1 (Lipschitz bound for Q_{arch}). *For any $K > 0$ we have*

$$|Q_{\text{arch}}(\Phi_1) - Q_{\text{arch}}(\Phi_2)| \leq L_{\text{arch}}(K) \|\Phi_1 - \Phi_2\|_\infty$$

for all $\Phi_1, \Phi_2 \in C_{c,\text{even}}^\infty K$, where

$$L_{\text{arch}}(K) := \int_{-K}^K |a^*(\xi)| d\xi < \infty. \tag{19}$$

Proof. Since Q_{arch} is linear and $\text{supp } \Phi \subset [-K, K]$ for $\Phi \in C_{c,\text{even}K}^\infty$,

$$|Q_{\text{arch}}(\Phi_1) - Q_{\text{arch}}(\Phi_2)| = \left| \int_{-K}^K a^*(\xi) (\Phi_1(\xi) - \Phi_2(\xi)) d\xi \right| \leq \int_{-K}^K |a^*(\xi)| d\xi \cdot \|\Phi_1 - \Phi_2\|_\infty.$$

The integral is finite because a^* is continuous on $[-K, K]$. $\square$

5.2 Lipschitz estimate for the prime part

We now turn to the prime contribution Q_{prime} . Recall that in the Guinand–Weil normalization the prime part of the Weil functional is given by

$$Q_{\text{prime}}(\Phi) = \sum_{\xi_n \in [-K, K]} w_Q(n) \Phi(\xi_n) = 2 \sum_{\substack{n \geq 2 \\ \xi_n \in [-K, K]}} \frac{\Lambda(n)}{\sqrt{n}} \Phi(\xi_n), \quad (20)$$

where $\xi_n = (\log n)/(2\pi)$ and Λ denotes the von Mangoldt function. Since $\Phi \in C_{c,\text{even}K}^\infty$ vanishes outside $[-K, K]$, this is a *finite* sum over those n with $\xi_n \in [-K, K]$, i.e., over $2 \leq n \leq e^{2\pi K}$.

Proposition 5.2 (Lipschitz bound for Q_{prime}). *For any $K > 0$ we have*

$$|Q_{\text{prime}}(\Phi_1) - Q_{\text{prime}}(\Phi_2)| \leq L_{\text{prime}}(K) \|\Phi_1 - \Phi_2\|_\infty$$

for all $\Phi_1, \Phi_2 \in C_{c,\text{even}K}^\infty$, where

$$L_{\text{prime}}(K) := 2 \sum_{2 \leq n \leq e^{2\pi K}} \frac{\Lambda(n)}{\sqrt{n}} < \infty. \quad (21)$$

Proof. Since Q_{prime} is linear and the sum is finite,

$$|Q_{\text{prime}}(\Phi_1) - Q_{\text{prime}}(\Phi_2)| = \left| 2 \sum_{2 \leq n \leq e^{2\pi K}} \frac{\Lambda(n)}{\sqrt{n}} (\Phi_1(\xi_n) - \Phi_2(\xi_n)) \right| \leq 2 \sum_{2 \leq n \leq e^{2\pi K}} \frac{\Lambda(n)}{\sqrt{n}} \cdot \|\Phi_1 - \Phi_2\|_\infty.$$

$\square$

Remark 5.3 (Explicit estimate for $L_{\text{prime}}(K)$). Using the trivial bound $\Lambda(n) \leq \log n$ and partial summation, one can show that

$$L_{\text{prime}}(K) \asymp \sqrt{e^{2\pi K}} = e^{\pi K}$$

as $K \rightarrow \infty$. More precisely, by the prime number theorem, $\sum_{n \leq x} \Lambda(n)/\sqrt{n} \sim 2\sqrt{x}$, so $L_{\text{prime}}(K) \sim 4e^{\pi K}$.

5.3 Main Lipschitz theorem

We can now combine the Archimedean and prime bounds to obtain the main Lipschitz estimate for Q on each compact frequency window.

Theorem 5.4 (Lipschitz continuity of Q). *Let $K > 0$. Then the Weil functional Q is Lipschitz continuous on the space $C_{c,\text{even}K}^\infty$ of admissible test functions, equipped with the supremum norm. Specifically,*

$$|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_\infty$$

for all $\Phi_1, \Phi_2 \in C_{c,\text{even}K}^\infty$, where the Lipschitz constant is

$$L_Q(K) = L_{\text{arch}}(K) + L_{\text{prime}}(K) = \int_{-K}^K |a^*(\xi)| d\xi + 2 \sum_{2 \leq n \leq e^{2\pi K}} \frac{\Lambda(n)}{\sqrt{n}}. \quad (22)$$

Proof. This is an immediate consequence of the decomposition $Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi)$ and the triangle inequality:

$$|Q(\Phi_1) - Q(\Phi_2)| \leq |Q_{\text{arch}}(\Phi_1) - Q_{\text{arch}}(\Phi_2)| + |Q_{\text{prime}}(\Phi_1) - Q_{\text{prime}}(\Phi_2)| \leq (L_{\text{arch}}(K) + L_{\text{prime}}(K)) \|\Phi_1 - \Phi_2\|_{\infty}.$$

□

Remark 5.5 (Growth of $L_Q(K)$). As $K \rightarrow \infty$, the Archimedean contribution $L_{\text{arch}}(K)$ grows like $O(K \log K)$ (from the logarithmic growth of a^*), while the prime contribution $L_{\text{prime}}(K)$ grows exponentially like $e^{\pi K}$. Thus for large K the Lipschitz constant is dominated by the prime part.

Remark 5.6 (Extension to continuous functions). The proof of theorem 5.4 does not use the smoothness of test functions, only their values on $[-K, K]$. Consequently, the Lipschitz bound extends verbatim to all continuous functions on $[-K, K]$: for any $\Phi_1, \Phi_2 \in C([-K, K])$ (extended by zero outside $[-K, K]$) one has $|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_{\infty}$. In particular, this applies to the Fejér–heat cone $\mathcal{G}_K$ from section 4, ensuring that positivity results on $\mathcal{G}_K$ propagate by density and continuity to the full cone $C_{\text{even}}^+([-K, K])$.

6 Discussion and outlook

In this note we have established two basic analytic tools for studying the Weil quadratic functional Q on compact frequency windows:

- the density of the Fejér–heat cone $\mathcal{G}_K$ in the cone $C_{\text{even}}^+([-K, K])$ (theorem 1.1, proved in theorem 4.5);
- the Lipschitz continuity of Q with an explicit constant $L_Q(K)$ on each window $[-K, K]$ (theorem 1.2, proved in theorem 5.4).

Both results are formulated and proved entirely within the Guinand–Weil normalization of the explicit formula for the Riemann zeta function, and they do not rely on any global positivity assumptions.

Connection to the operator–theoretic framework

These results may serve as building blocks for operator–theoretic approaches to the Weil positivity criterion. On each compact window $[-K, K]$ the Fejér–heat windows are particularly well suited to two complementary roles:

- *Toeplitz operators.* The Archimedean contribution Q_{arch} can be represented as a Toeplitz form with a smooth symbol arising from the gamma factor (see [2] for general background on Toeplitz operators). The Fejér–heat generators provide a convenient class of test windows for which lower bounds on Toeplitz eigenvalues can be accessed via classical symbol estimates.
- *Reproducing kernel Hilbert spaces.* The heat kernel ρ_t generates a reproducing kernel Hilbert space (RKHS) on the frequency side (see [1, 10] for RKHS theory). Embedding the prime contribution Q_{prime} into such an RKHS makes it possible to bound the corresponding “prime operator” by explicit RKHS norms, with Gaussian damping playing a key role in the tail estimates.

The Lipschitz estimate $L_Q(K)$ ensures that positivity results for Q established on dense subsets of the frequency cones extend by continuity to the full cones. In particular, if one could prove that $Q(\Phi) \geq 0$ for all Φ in a dense subcone of $C_{\text{even}}^+([-K, K])$ generated by Fejér–heat windows, then theorem 1.1 and theorem 1.2 would allow one to propagate this to all admissible test functions on $[-K, K]$.

Relation to the Riemann Hypothesis

The Weil positivity criterion states that

$$\text{Riemann Hypothesis} \iff Q(\Phi) \geq 0 \quad \text{for all admissible } \Phi,$$

with Q defined in the Guinand–Weil normalization (see [7, 11, 12]). The present work does not attempt to establish any such global positivity statement. This note is purely technical in nature: it isolates density and continuity statements that may be used as components of such arguments, independently of their eventual success.

In a separate work, we combine the tools developed here with:

- lower bounds on the minimal eigenvalue of Toeplitz operators arising from the Archimedean symbol;
- RKHS norm estimates for the prime operator, based on Gaussian kernels and reproducing properties;
- a compact-by-compact transfer mechanism that passes from positivity on each window $[-K, K]$ to positivity on the full Weil cone of admissible test functions.

Taken together, these ingredients suggest an operator-theoretic framework for studying the Weil positivity criterion. Regardless of the eventual outcome of any broader program along these lines, the Fejér–heat density and Lipschitz control results proved in this note are self-contained analytic statements about the localized Weil functional and may be of independent interest in the study of explicit formulas and quadratic forms attached to zeta and L -functions.

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